Overhead Crane Inspection Checklist (29 CFR 1910.179)

For frequent and periodic crane inspections, monthly hook, chain and rope checks, and cranes returning from idle.

OSHA 29 CFR 1910.179

DATE	SHIFT
INSPECTOR	AREA / EQUIPMENT ID

Before you start

1. Mark every item OK, Defect or N/A.
2. If an item is marked Defect, decide whether it is a safety hazard and keep the crane out of operation until designated personnel correct any unsafe condition (1910.179(l)(3)(i)).
3. The crane manufacturer's instructions and your site rules may add checks to this list.

Legal basis: 29 CFR 1910.179: Overhead and gantry cranes (Part 1910 Subpart N, Materials Handling and Storage) Source: eCFR, Title 29
Online version and sources: https://safetyindex.blog/us/standards/1910-179-overhead-gantry-cranes

SECTION A | Frequent inspection items

No.	Check item	What to look for	OK	Defect	N/A	Notes
1	Operating mechanisms (daily) Ref. 1910.179(j)(2)(i)	All functional operating mechanisms work with no maladjustment interfering with proper operation	☐	☐	☐	
2	Air and hydraulic systems (daily) Ref. 1910.179(j)(2)(ii)	No deterioration or leakage in lines, tanks, valves, drain pumps or other air or hydraulic parts	☐	☐	☐	
3	Hooks Ref. 1910.179(j)(2)(iii)	No cracks or deformation; throat opening no more than 15 percent over normal; twist no more than 10° from the plane of the unbent hook	☐	☐	☐	
4	Hook monthly record Ref. 1910.179(j)(2)(iii)	Certification record for the latest monthly inspection with date, inspector's signature and hook serial number or other identifier	☐	☐	☐	
5	Hoist chains and end connections Ref. 1910.179(j)(2)(iv)	No excessive wear, twist or distorted links interfering with function; stretch within manufacturer limits	☐	☐	☐	
6	Hoist chain monthly record Ref. 1910.179(j)(2)(iv)	Monthly record with date, signature and chain identifier	☐	☐	☐	
7	Mechanism wear Ref. 1910.179(j)(2)(vi)	No excessive wear of components in functional operating mechanisms	☐	☐	☐	
8	Rope reeving Ref. 1910.179(j)(2)(vii)	Reeving matches the crane manufacturer's recommendations	☐	☐	☐	

SECTION B | Periodic inspection items (check section A too)

No.	Check item	What to look for	OK	Defect	N/A	Notes
9	Structural members Ref. 1910.179(j)(3)(i)	No deformed, cracked or corroded members	☐	☐	☐	
10	Bolts and rivets Ref. 1910.179(j)(3)(ii)	No loose bolts or rivets	☐	☐	☐	
11	Sheaves and drums Ref. 1910.179(j)(3)(iii)	No cracked or worn sheaves or drums	☐	☐	☐	

General guidance only, not legal advice. Check the current regulation and your employer's procedures.

SECTION B | Periodic inspection items (check section A too)

No.	Check item	What to look for	OK	Defect	N/A	Notes
12	Pins, bearings, shafts and gears Ref. 1910.179(j)(3)(iv)	No worn, cracked or distorted pins, bearings, shafts, gears, rollers, locking or clamping devices	☐	☐	☐	
13	Brake system Ref. 1910.179(j)(3)(v)	No excessive wear on brake parts, linings, pawls or ratchets	☐	☐	☐	
14	Load, wind and other indicators Ref. 1910.179(j)(3)(vi)	No significant inaccuracy over the full range	☐	☐	☐	
15	Powerplants Ref. 1910.179(j)(3)(vii)	Gasoline, diesel, electric or other powerplant performs properly and meets applicable safety requirements	☐	☐	☐	
16	Chain drive sprockets Ref. 1910.179(j)(3)(viii)	No excessive sprocket wear or chain stretch	☐	☐	☐	
17	Electrical apparatus Ref. 1910.179(j)(3)(x)	No pitting or deterioration of controller contactors, limit switches or pushbutton stations	☐	☐	☐	

SECTION C | Running ropes and drum

No.	Check item	What to look for	OK	Defect	N/A	Notes
18	Rope diameter Ref. 1910.179(m)(1)(i)	Not reduced below nominal by loss of core support, corrosion or wear of outside wires	☐	☐	☐	
19	Broken outside wires Ref. 1910.179(m)(1)(ii)	Number and concentration of broken outside wires checked and judged not to be a hazard	☐	☐	☐	
20	Worn outside wires Ref. 1910.179(m)(1)(iii)	No worn outside wires causing appreciable loss of strength	☐	☐	☐	
21	Wires at end connections Ref. 1910.179(m)(1)(iv)	No corroded or broken wires at end connections	☐	☐	☐	
22	End connections Ref. 1910.179(m)(1)(v)	No corroded, cracked, bent, worn or improperly applied end connections	☐	☐	☐	
23	Kinks and damage Ref. 1910.179(m)(1)(vi)	No severe kinking, crushing, cutting or unstranding	☐	☐	☐	
24	Wraps on drum Ref. 1910.179(h)(2)(iii)(a)	At least two wraps remain on the drum with the hook at its extreme low position	☐	☐	☐	
25	Monthly rope record Ref. 1910.179(m)(1)	Record on file showing date, signature and the ropes inspected	☐	☐	☐	

SECTION D | Markings, controls and safety devices

No.	Check item	What to look for	OK	Defect	N/A	Notes
26	Rated load marking Ref. 1910.179(b)(5)	Rated load plainly marked on each side of the crane; each hoist's rated load on the hoist or load block, legible from the floor	☐	☐	☐	
27	Pendant control box Ref. 1910.179(g)(1)(v)	Box protects against electrical shock; every function clearly marked	☐	☐	☐	
28	Pendant station condition Ref. 1910.179(l)(3)(iii)(d)	Station clean; function labels legible	☐	☐	☐	
29	Pendant pushbuttons Ref. 1910.179(g)(3)(vii)	Each button returns to off when pressure is released	☐	☐	☐	

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SECTION D | Markings, controls and safety devices

No.	Check item	What to look for	OK	Defect	N/A	Notes
30	Upper limit switch (start of shift) Ref. 1910.179(n)(4)(i)	Tried under no load with the block inched in or run at slow speed; switch operates	☐	☐	☐	
31	Warning signal Ref. 1910.179(i)	Gong or other effective warning signal works (not required on floor-operated cranes)	☐	☐	☐	
32	Runway power disconnect Ref. 1910.179(g)(5)(i)	Switch or circuit breaker on a fixed structure, reachable from the floor, can be locked open	☐	☐	☐	
33	Moving-part guards Ref. 1910.179(e)(6)(i)	Exposed gears, set screws, projecting keys, chains and sprockets are guarded	☐	☐	☐	
34	Trolley stops Ref. 1910.179(e)(1)(i)	Stops in place at the limits of trolley travel	☐	☐	☐	
35	Rail sweeps Ref. 1910.179(e)(4)	Bridge trucks have sweeps extending below the top of the rail and projecting in front of the wheels	☐	☐	☐	

DEFECTS FOUND AND ACTION TAKEN

Item no.	Defect description	Reported to	Action taken / date closed

SIGN-OFF

SAFE TO OPERATE? ☐ Yes ☐ No: tagged out of service	
INSPECTOR SIGNATURE	DATE / TIME
SUPERVISOR REVIEW	DATE / TIME